

Bidirectional LSTM Implementation-24/10/2025(FRIDAY)

Objective

The goal of this project is to develop a **Bidirectional LSTM (BiLSTM) based language model** that can understand and generate human-like text using the **NLTK Gutenberg corpus**. The task focuses on **next-word prediction** to demonstrate natural language processing (NLP) capabilities.

Tools and Libraries

- **Google Colab** (development environment)
- **Python 3.12, TensorFlow 2.19, Keras, NLTK, NumPy, Matplotlib**

Dataset

- The **Gutenberg corpus** from NLTK, specifically three **Jane Austen** novels: *Emma*, *Persuasion*, and *Sense and Sensibility*.

Methodology

1. **Data Preprocessing:**
 - Lowercasing, cleaning punctuation, and removing unwanted characters.
 - Tokenization and lemmatization using **WordNetLemmatizer**.
2. **Vocabulary Building:**
 - Tokens converted to integer sequences using Keras **Tokenizer**.
3. **Sequence Generation:**
 - Sliding windows of 20 words used to predict the next word.
4. **Model Design:**
 - Embedding → Bidirectional LSTM (256) → Dense (256, ReLU) → Softmax Output
5. **Training:**
 - Optimizer: Adam (3e-4)
 - Loss: Sparse Categorical Cross-Entropy
 - Callbacks: Early Stopping and ModelCheckpoint
6. **Evaluation Metrics:**
 - Accuracy and **Perplexity** on validation data.

Results

- The model achieved good validation accuracy and moderate perplexity, indicating effective next-word prediction capability.
- Generated text samples were grammatically coherent and stylistically close to Austen's writing.

Conclusion

This project demonstrates how **Bidirectional LSTMs** can effectively capture **context from both directions** in text, improving prediction quality over unidirectional models. Using the **Gutenberg corpus** highlights how deep learning can model classical English literature and perform meaningful text generation.

SUMMARY:

This project builds a **Bidirectional LSTM language model** using the **NLTK Gutenberg corpus** to perform **next-word prediction**. After cleaning and tokenizing text from Jane Austen's novels, the data was trained on a BiLSTM network. The model successfully learned word dependencies and generated realistic text sequences. It demonstrates the application of deep learning and NLP techniques in understanding and generating human-like language.